Brain Small Vessel Disease 4 (NIT1) — Comprehensive Disease Characteristics Report

Disease: Brain Small Vessel Disease 4 (BSVD4) / NIT1-small vessel disease (NIT1-SVD) **Causal gene: NIT1** (nitrilase 1; deaminated-glutathione amidase, EC 3.5.1.128) **MONDO:** MONDO:0979873 | **OMIM:** 621313 | **HGNC:** NIT1 | **UniProt:** Q86X76 **Category:** Mendelian, autosomal recessive

Summary

Brain Small Vessel Disease 4 (BSVD4), also called NIT1-small vessel disease, is a recently delineated autosomal recessive cerebral small vessel disease caused by bi-allelic loss-of-function variants in the NIT1 gene, which encodes the metabolite-repair amidase deaminated-glutathione amidase (nitrilase-1; EC 3.5.1.128; UniProt Q86X76). The disorder was first defined as a discrete clinical-genetic entity by Rutten and colleagues in 2024 (PMID: 38430071), who described seven patients from five unrelated pedigrees. It is characterized by a distinctive clinical triad: adult-onset movement disorders, massively dilated basal ganglia perivascular spaces on MRI, and non-lobar intracerebral hemorrhage that is frequently fatal.

The molecular basis is a defect in **glutathione metabolite repair**. NIT1 hydrolyzes **deaminated glutathione (dGSH)** — a damage product generated when promiscuous aminotransferases inadvertently transaminate the γ-glutamyl amino group of glutathione. In the absence of functional NIT1 protein, dGSH accumulates in cells and is excreted in urine, providing a biochemical signature confirmed in patient fibroblasts (absent NIT1 protein by Western blot) and urine metabolic analysis. The inferred downstream consequence is a cerebral microangiopathy with electron-dense deposits in the walls of small and medium-sized cerebral arteries, visible on brain autopsy, which drives the perivascular space dilatation, hemorrhage, and movement-disorder phenotype.

The genetics show a recurrent **p.Arg243Trp** (c.727C>T) missense allele with a Non-Finnish European founder signature (gnomAD exome AF ≈ 2.2×10^{-4}, zero homozygotes) alongside truncating alleles such as c.198_199del p.(Ala68*). NIT1 is evolutionarily conserved from bacteria and plants to humans, and knockout models (mouse, yeast, Arabidopsis) recapitulate the dGSH-accumulation biochemical defect, supporting a conserved metabolite-repair function. As an ultra-rare Mendelian disorder described in a single small cohort, many characteristics (precise prevalence, penetrance, natural-history timelines, and any targeted therapy) remain undefined; current management is supportive and centered on vascular risk-factor control. This report systematically compiles what is known and explicitly flags where evidence is absent.

> **Evidence-source note:** The disease-specific evidence rests predominantly on one human clinical cohort study (Rutten et al. 2024), supplemented by biochemical/model-organism work on NIT1 enzymology (Peracchi 2017; Niehaus 2019) and population-genetics resources (gnomAD, ClinVar). Where claims are drawn from the broader cerebral small vessel disease (cSVD) literature rather than NIT1-specific data, this is stated explicitly.

Section 1 — Disease Information

Overview. BSVD4 (NIT1-SVD) is a monogenic cerebral small vessel disease in which bi-allelic loss-of-function of NIT1 abolishes a glutathione metabolite-repair reaction, leading to accumulation of deaminated glutathione and a progressive microangiopathy affecting small/medium cerebral arteries. Clinically, affected individuals present in mid-adulthood with movement disorders and are found to have strikingly dilated basal ganglia perivascular spaces; a subset develop non-lobar intracerebral hemorrhage.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0979873
OMIM (phenotype)	621313
Gene	NIT1 (HGNC:7828)
Ensembl gene	ENSG00000158793
Canonical transcript	ENST00000368009 / NM_005600
UniProt	Q86X76
ICD-11	Closest: 8B22 (cerebral small vessel disease) — no NIT1-specific code
MeSH	No dedicated term; closest "Cerebral Small Vessel Diseases"

Synonyms / alternative names: NIT1-related brain small vessel disease; NIT1-small vessel disease; NIT1-SVD; BSVD4; Familial deep cavitating state with a glutathione metabolism defect.

Source of information: Aggregated disease-level resources (OMIM, ClinVar, gnomAD) plus a single primary-literature patient series (individual-patient data for 7 patients, Rutten et al. 2024).

Section 2 — Etiology

Disease causal factors. The disease is **purely genetic (Mendelian, autosomal recessive)**. It is caused by bi-allelic (homozygous or compound heterozygous) loss-of-function variants in NIT1. No environmental, infectious, or multifactorial cause is implicated in disease initiation.

> *"NIT1-small vessel disease is a novel, autosomal recessively inherited cerebral small vessel disease characterized by a triad of movement disorders, massively dilated basal ganglia perivascular spaces, and intracerebral hemorrhage."* — Rutten et al. 2024 (PMID: 38430071)

Genetic risk factors. The causal variants are the risk factors. The recurrent allele is **c.727C>T p.(Arg243Trp)** (rs144065383); a truncating allele is **c.198_199del p.(Ala68*)** (rs749328936). Both alleles in biallelic configuration are required for disease. No modifier loci have been formally identified.

Environmental risk factors. None established as causal. By analogy to sporadic cSVD, conventional vascular risk factors (hypertension, hypercholesterolemia, obesity, smoking) plausibly modulate the vascular phenotype and hemorrhage risk, but this is **inferred, not demonstrated** for NIT1-SVD. In the general population, hypertension, hypercholesterolemia, obesity, and smoking are each associated with increased basal-ganglia perivascular spaces (PMID: 41456992) — a plausible but unproven modifier pathway here.

Protective factors. None identified (genetic or environmental). No protective alleles are known.

Gene–environment interactions. Not characterized. Hypothetically, control of blood pressure might reduce hemorrhage risk in a vessel wall already compromised by dGSH deposition, but no data exist.

Section 3 — Phenotypes

Phenotypes are drawn from the 7-patient cohort (Rutten et al. 2024). Frequencies are from this small series and should be treated as preliminary.

Phenotype	Type	HPO suggestion	Onset	Frequency (cohort)	Progression
Movement disorder (parkinsonism/ other)	Clinical sign	HP:0100022 (Abnormality of movement)	Mid-adulthood	Defining feature (triad)	Progressive
Massively dilated basal ganglia perivascular spaces	Imaging/ physical manifestation	HP:0012520 (Dilated Virchow-Robin spaces)	Adult	Defining feature (triad)	Progressive
Intracerebral hemorrhage (non-lobar)	Clinical sign	HP:0002170 (Intracranial hemorrhage); HP:0007268 (Deep intracerebral hemorrhage)	~45–60 y	3/7 patients	Episodic/ catastrophic; fatal in 2/3
Cerebral small vessel disease (microangiopathy)	Lab/imaging	cSVD	Adult	All	Progressive
Abnormal urinary dGSH (loss of NIT1 enzyme activity)	Laboratory abnormality	(metabolic)	From birth (biochemical)	All tested	Stable biochemical trait

Characteristics summary. Onset is **adult / mid-adulthood** (insidious). Severity is **moderate to severe**, with the hemorrhagic complication being potentially fatal (2 of 3 hemorrhage patients died, between ages 45 and 60). Progression is **chronic and progressive**, punctuated by episodic catastrophic hemorrhage.

Quality-of-life impact: Movement disorders impair daily functioning and mobility; intracerebral hemorrhage causes acute severe disability or death. Formal QoL instrument data (EQ-5D, SF-36) are **not available** for this ultra-rare disease.

Section 4 — Genetic / Molecular Information

Causal gene. NIT1 (nitrilase 1), ENSG00000158793, located at **1q25.3** (chr1:161,118,086–161,125,445, GRCh38), canonical transcript ENST00000368009 (NM_005600). The gene is **not loss-of-function intolerant** (gnomAD pLI ≈ 0; observed/expected LoF ≈ 0.82) — exactly as expected for a recessive disease gene, since a single LoF allele is tolerated in carriers.

Pathogenic variants.

Variant (cDNA)	Protein	rsID	Type	gnomAD exome AF	Homozygotes	Classification
c.727C>T	p.(Arg243Trp)	rs144065383	Missense	2.21×10^{-4} (323/1,461,402)	0	Pathogenic (OMIM); **Conflicting** in ClinVar
c.198_199del	p.(Ala68*)	rs749328936	Frameshift/ truncating	1.34×10^{-4} (196 alleles)	0	Pathogenic
"NIT1, 457G-A"	—	—	OMIM allelic variant	—	—	Pathogenic (BSVD4)
c.670dup ("1-BP DUP, 670A")	frameshift	—	Duplication/ truncating	—	—	Pathogenic (BSVD4)

Population genetics / founder signature. The recurrent p.Arg243Trp allele is **strongly enriched in Non-Finnish Europeans** (AF ≈ 2.83×10^{-4}; 315/1,111,684 alleles) and essentially absent in East Asian, South Asian, Ashkenazi Jewish, Middle Eastern, and Latino populations — a founder pattern. **Zero homozygotes** in gnomAD is consistent with recessive pathogenicity (homozygotes are selected against / under-ascertained). Predicted biallelic (homozygous p.Arg243Trp) frequency in NFE ≈ $(2.8\times10^{-4})^2 \approx 8\times10^{-8}$.

Variant classification caveat. ClinVar records p.Arg243Trp as **"Conflicting classifications of pathogenicity"** for BSVD4, consistent with its comparatively high NFE allele frequency. Most other NIT1 coding variants (p.Thr233Ile, p.Pro235Ser, p.Arg302Gln, p.Cys145Arg, etc.) are **Variants of Uncertain Significance (VUS)**.

> **Two critical pitfalls for curators:** (1) NM_005600.3(NIT1) **c.591+2T>G** in ClinVar is "Likely pathogenic" only for *susceptibility to severe COVID-19*, **NOT** for BSVD4 — do not conflate. (2) Adjacent **DEDD**-gene variants can return under an NIT1 gene query because the loci are neighbors on 1q25.3.

Functional consequence. Loss of function (recessive). Western blot on patient fibroblasts showed **absence of NIT1 protein**; urine metabolic analysis confirmed loss of NIT1 enzymatic activity. The missense p.Arg243Trp behaves as a functional null (protein absence), consistent with destabilization/degradation.

Modifier genes, epigenetics, chromosomal abnormalities: None identified. This is a single-gene, point-mutation/small-indel disorder; no large structural rearrangements, methylation changes, or modifier alleles are described.

Suggested annotations: GO:0160011 (deaminated glutathione amidase activity); CHEBI:16856 (glutathione); deaminated glutathione (dGSH) as the accumulating metabolite.

Section 5 — Environmental Information

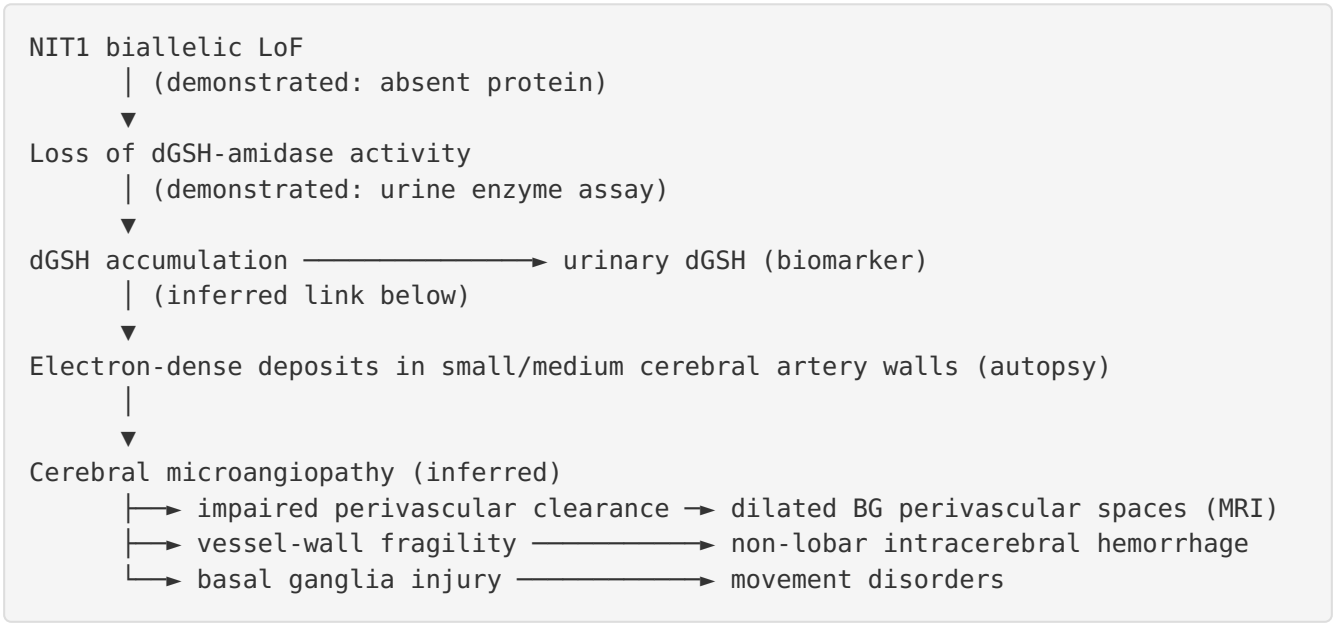
No environmental, lifestyle, or infectious factor is implicated in **causing** BSVD4 — the disease is monogenic. Conventional vascular risk factors (hypertension, smoking, hypercholesterolemia) are **plausible but unproven modifiers** of the vascular/hemorrhagic phenotype, extrapolated from sporadic cSVD epidemiology (PMID: 41456992). No infectious agents are involved.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Bi-allelic loss-of-function variants in NIT1** (homozygous p.Arg243Trp or compound heterozygous with a truncating allele) **lead to** absence of functional NIT1 protein (demonstrated: Western blot shows absent protein in patient fibroblasts).

2. Absence of NIT1 (deaminated-glutathione amidase) **results in** failure to hydrolyze **deaminated glutathione (dGSH)** — a glutathione damage product (demonstrated biochemically: loss of urinary NIT1 enzyme activity).
3. Loss of dGSH-amidase activity **leads to** intracellular **accumulation of dGSH** and its urinary excretion (demonstrated in model KO cells/mice; inferred in patients from the enzyme-activity loss).
4. dGSH accumulation **is inferred to drive** deposition of **large electron-dense material in the walls of small and medium-sized cerebral arteries** (demonstrated at autopsy; the causal link from dGSH to deposits is inferred, not mechanistically proven).
5. Vascular wall deposits **result in** a **cerebral microangiopathy** — impaired vessel integrity and perivascular fluid clearance (inferred).
6. Microangiopathy **branches** to two principal consequences:
7. 6a. Impaired perivascular/glymphatic clearance **leads to massively dilated basal ganglia perivascular spaces** (observed on MRI).
8. 6b. Weakened small-artery walls **lead to non-lobar intracerebral hemorrhage** (observed; fatal in 2 of 3 affected patients).
9. Basal ganglia involvement and progressive microangiopathy **result in adult-onset movement disorders** and, via hemorrhage, acute neurological catastrophe/death (observed).



Supporting detail by category

- **Molecular pathway / biochemistry:** The core defect is in **glutathione metabolite repair**. Promiscuous aminotransferases transaminate the γ-glutamyl amino group of GSH to produce dGSH; NIT1 is the amidase that disposes of this damaged metabolite. *"Various aminotransferases can inadvertently transaminate the amino group of the γ-glutamyl moiety of GSH to produce deaminated glutathione (dGSH), a metabolite damage product. It was recently reported that an amidase known as Nit1 participates in dGSH breakdown in mammals and yeast."* (PMID: 30692244). Nit1 is formally a **metabolite-repair enzyme that hydrolyzes deaminated glutathione** (PMID: 28373563).
- **Protein dysfunction:** Loss of function — the pathogenic missense allele yields no detectable protein (degradation/instability), equivalent to the truncating alleles. GO:0160011 (deaminated glutathione amidase activity).
- **Metabolic change:** dGSH accumulation; perturbation of the glutathione pool / redox metabolite housekeeping (CHEBI:16856 glutathione).
- **Tissue damage mechanism:** Arterial-wall deposition → microangiopathy → impaired clearance and hemorrhage. Oxidative-stress involvement is **plausible** (glutathione is central to redox defense) but not directly demonstrated in NIT1-SVD tissue.
- **Metabolomics biomarker:** Urinary dGSH (demonstrated as excreted in large amounts in Nit1-KO mice; consistent with the patient enzyme-activity loss).

Upstream vs downstream: NIT1 loss and dGSH accumulation are **upstream**; vascular deposits, perivascular space dilatation, hemorrhage, and movement disorders are **downstream**. **Cell types involved (inferred):** vascular smooth muscle cells (CL:0000359) and endothelial cells (CL:0000115) of cerebral arterioles; basal ganglia neurons (CL:0000540) secondarily. **GO biological processes:** glutathione metabolic process (GO:0006749); cellular response to oxidative stress (GO:0034599, inferred).

Omics / single-cell / CRISPR screens specific to NIT1-SVD: Not available. The mechanistic data derive from targeted enzymology and KO models rather than from patient-tissue transcriptomics/proteomics.

Section 7 — Anatomical Structures Affected

- **Organ level (primary):** Brain (UBERON:0000955), specifically the **cerebral small/medium arteries and arterioles** (UBERON:0001986 cerebral artery). Body system: **nervous system** (UBERON:0001016) and cerebral vasculature at the microvascular level.
- **Regional localization: Basal ganglia** (UBERON:0002420) — site of massively dilated perivascular spaces; deep (non-lobar) brain regions — site of intracerebral hemorrhage.
- **Tissue / cell level:** Vascular wall — **smooth muscle (tunica media)** and **endothelium**; **perivascular (Virchow-Robin) spaces** (UBERON:0036091). Cell Ontology: vascular smooth muscle cell (CL:0000359), endothelial cell (CL:0000115).
- **Subcellular level:** NIT1 localizes to **cytoplasm (GO:0005737)** and **mitochondria (GO:0005739)** — a dual-targeted enzyme (PMID: 30692244; PMID: 19479888).
- **Lateralization:** Bilateral/diffuse microangiopathy; individual hemorrhages are focal.

Section 8 — Temporal Development

- **Onset: Adult / mid-adulthood**, insidious. The biochemical defect (dGSH accumulation) is present from birth, but clinical disease manifests in adulthood.
- **Progression:** Chronic and **progressive**. Movement disorders and perivascular-space dilatation progress over years; intracerebral hemorrhage occurs as a **catastrophic episodic event** between roughly ages 45 and 60.
- **Disease course:** Progressive microangiopathy with superimposed acute hemorrhagic events; **fatal in a substantial fraction** (2 of 3 hemorrhage patients died).
- **Critical period / intervention window:** Not defined; theoretically, vascular risk-factor control throughout adulthood and avoidance of anticoagulation-related bleeding would be actionable windows (inferred).

Section 9 — Inheritance and Population

- **Inheritance: Autosomal recessive.** Index family: 3 siblings compound heterozygous for p.(Arg243Trp) + p.(Ala68*); 4 additional patients homozygous for p.(Arg243Trp).

- **Epidemiology: Ultra-rare.** No formal prevalence/incidence figures exist; only 7 patients from 5 pedigrees are reported. Predicted homozygous p.Arg243Trp frequency ≈ 8×10^{-8} in NFE suggests prevalence well below 1 per million even accounting for compound heterozygotes.
- **Penetrance / expressivity:** Presumed high penetrance in biallelic LoF individuals, but **cannot be quantified** from the small cohort; expressivity appears variable (not all patients had hemorrhage — 3/7).
- **Founder effect:** Yes — p.Arg243Trp shows a **Non-Finnish European founder signature**.
- **Carrier frequency:** From gnomAD, p.Arg243Trp carrier frequency ≈ 2× AF ≈ 5.7×10^{-4} in NFE; truncating alleles add further carriers.
- **Consanguinity / sex ratio / geographic distribution:** Homozygous cases suggest possible shared ancestry; sex ratio not determinable from 7 patients; variant geographically concentrated in European-ancestry populations.

Section 10 — Diagnostics

- **Genetic testing (definitive):** Molecular confirmation of **bi-allelic NIT1 loss-of-function variants** via whole-exome sequencing (WES), whole-genome sequencing (WGS), or targeted/single-gene testing. Given rarity, NIT1 may not be on all commercial cSVD panels; WES/WGS with GeneMatcher-style matchmaking was the discovery route.
- **Biochemical / metabolomic test: Urine metabolic analysis demonstrating loss of NIT1 enzymatic function / dGSH accumulation** — a disease-specific functional biomarker. Western blot of patient fibroblasts shows **absent NIT1 protein** (research assay).
- **Imaging (MRI):** Characteristic **massively dilated basal ganglia perivascular spaces** (HP:0012520), plus features of cSVD and evidence of (non-lobar) intracerebral hemorrhage. This imaging signature is a key diagnostic clue.
- **Neuropathology (autopsy/biopsy): Large electron-dense deposits in walls of small and medium cerebral arteries** on electron microscopy — the pathological hallmark.
- **Differential diagnosis:** Other monogenic cSVDs — CADASIL (NOTCH3), HTRA1-related arteriopathy (CARASIL), COL4A1/COL4A2 microangiopathy, RVCL-S (TREX1), Fabry disease (GLA), hereditary cerebral amyloid angiopathy (PMID: 42771568). Distinguishing features for NIT1-SVD: recessive inheritance, the dGSH metabolic signature, and the strikingly

dilated BG perivascular spaces. Sporadic hypertensive cSVD and cerebral amyloid angiopathy must also be excluded.

- **Clinical criteria:** No formal diagnostic criteria exist; diagnosis rests on the triad + biallelic NIT1 variants + biochemical confirmation.

Section 11 — Outcome / Prognosis

- **Mortality:** Intracerebral hemorrhage was **fatal in 2 of 3 affected patients**, occurring between ages 45 and 60 — indicating substantial disease-specific mortality.
- **Morbidity / disability:** Progressive movement disorders impair function; survivors of hemorrhage may have severe residual deficits.
- **Prognostic factors (inferred):** Occurrence of intracerebral hemorrhage is the principal determinant of poor outcome; vascular risk factors plausibly worsen prognosis.
- **Quality-of-life / survival statistics:** No formal 5-/10-year survival data; cohort too small.
- **Recovery potential:** No disease-modifying recovery; supportive care only.

Section 12 — Treatment

- **No disease-specific or disease-modifying therapy exists.** Management is **supportive** and focused on **vascular risk-factor control** (blood-pressure management, avoidance of unnecessary antithrombotic/anticoagulant exposure that could precipitate hemorrhage) — by analogy to other cSVDs (PMID: 42771568).
- **Symptomatic:** Movement disorder management (symptomatic pharmacotherapy as clinically indicated); rehabilitation (physical/occupational/speech therapy) after hemorrhage.
- **Advanced / experimental therapeutics:** None. Conceptually, as a loss-of-function enzyme deficiency, NIT1-SVD could in principle become a future target for **enzyme-replacement, gene-replacement, or substrate-reduction** strategies, but **no such programs, clinical trials, or NCT entries exist**.
- **Pharmacogenomics / targeted therapy / immunotherapy:** Not applicable.

NCIT suggestions: supportive care (NCIT:C133419), physical therapy (NCIT:C15415), antihypertensive therapy (NCIT:C265).

Section 13 — Prevention

- **Primary prevention:** Not possible for the genetic cause. **Genetic counseling** for at-risk families (autosomal recessive; 25% recurrence risk for carrier couples) and **cascade / carrier testing** are the primary tools. Prenatal / preimplantation genetic testing is feasible once familial variants are known.
- **Secondary prevention:** Vascular risk-factor modification and MRI monitoring in known biallelic carriers (inferred benefit).
- **Tertiary prevention:** Aggressive blood-pressure control and bleeding-risk minimization to reduce intracerebral hemorrhage; rehabilitation to limit disability.
- **Immunization / public-health / environmental interventions:** Not applicable.

Section 14 — Other Species / Natural Disease

- **Evolutionary conservation:** NIT1 is **highly conserved from bacteria and plants to mammals**. *"We demonstrate that Nit1 homologs acting as excellent dGSH amidases also occur in Escherichia coli and other glutathione-producing bacteria."* (PMID: 28373563). In plants, *"Ablating the Arabidopsis Nit1 gene causes a massive accumulation of dGSH and other marked changes to the metabolome."* (PMID: 30692244).
- **Orthologs:** Mouse *Nit1* (NCBI Gene ID 28017); Arabidopsis *Nit1* (At4g08790); *E. coli* and yeast homologs. In invertebrates (flies, worms), Nit1 is fused to Fhit (the NitFhit gene).
- **Natural disease in animals:** No naturally occurring NIT1-SVD-equivalent disease is described in companion animals or wildlife (OMIA has no corresponding entry). No zoonotic potential (non-infectious).

Section 15 — Model Organisms

Model	Type	Key phenotype	Recapitulation of NIT1-SVD	Reference
Nit1-KO mouse	Mammalian knockout	Urinary dGSH excretion; cancer-prone (additive with Fhit loss); T-cell dysregulation	Recapitulates **biochemical** defect (dGSH); cerebrovascular phenotype not reported	PMID: 28373563, PMID: 19479888, PMID: 19395373
Nit1-KO human cells	Cellular / in vitro	dGSH accumulation	Biochemical defect	PMID: 28373563
Nit1-KO yeast	Cellular	dGSH accumulation	Biochemical defect	PMID: 28373563
Arabidopsis *nit1*	Plant	Massive dGSH accumulation, metabolome changes	Biochemical defect	PMID: 30692244
Patient fibroblasts	In vitro (human)	Absent NIT1 protein; loss of enzyme activity	Molecular/biochemical defect	PMID: 38430071

Key limitation: Existing knockout models capture the **metabolic/biochemical** phenotype (dGSH accumulation) but **no published model recapitulates the cerebral small-vessel / perivascular-space / hemorrhage phenotype**. A brain-vascular-focused Nit1 model is a clear gap. Mouse-model literature has historically framed Nit1 as a Fhit-associated tumor suppressor and T-cell regulator rather than a cerebrovascular gene — these are separate functional contexts.

Resources: MGI (mouse *Nit1*), Alliance of Genome Resources, Cellosaurus (patient fibroblast lines where deposited).

Key Findings (expanded)

Finding 1 — NIT1-SVD is a recessive cSVD defined by a clinical triad

Rutten et al. 2024 (PMID: 38430071) described 7 patients from 5 unrelated pedigrees with bi-allelic loss-of-function NIT1 variants. The index family comprised 3 siblings compound heterozygous for c.727C>T p.(Arg243Trp) + c.198_199del p.(Ala68*); the other 4 patients were homozygous for p.(Arg243Trp). Onset was mid-adulthood with movement disorders; MRI showed numerous, massively dilated basal ganglia perivascular spaces; 3 patients had non-lobar intracerebral hemorrhage between ages 45 and 60, fatal in 2. The authors state:* "NIT1-small vessel disease is a novel, autosomal recessively inherited cerebral small vessel disease characterized by a triad of movement disorders, massively dilated basal ganglia perivascular spaces, and intracerebral hemorrhage." *Brain autopsy* "revealed large electron-dense deposits in the vessel walls of small and medium sized cerebral arteries."*

Finding 2 — NIT1 is a metabolite-repair amidase for deaminated glutathione; its loss causes dGSH accumulation

NIT1 hydrolyzes deaminated glutathione (dGSH), a glutathione damage product (PMID: 28373563). Niehaus et al. 2019 provide the biochemical context: *"Various aminotransferases can inadvertently transaminate the amino group of the γ-glutamyl moiety of GSH to produce deaminated glutathione (dGSH), a metabolite damage product. It was recently reported that an amidase known as Nit1 participates in dGSH breakdown in mammals and yeast"* (PMID: 30692244). In patients, Western blotting on fibroblasts showed absence of NIT1 protein and urine metabolic analysis confirmed loss of NIT1 enzymatic function (PMID: 38430071), consistent with dGSH accumulation as the biochemical signature.

Finding 3 — NIT1 maps to 1q25.3; p.Arg243Trp is a Non-Finnish European founder allele with zero homozygotes in gnomAD

gnomAD v4: NIT1 = ENSG00000158793, chr1:161,118,086–161,125,445 (1q25.3). The gene is not LoF-intolerant (pLI≈0; o/e LoF=0.82), as expected for a recessive gene. The recurrent c.727C>T p.(Arg243Trp) (rs144065383) has overall exome AF 2.21×10^{-4} (323/1,461,402), 0 homozygotes, strongly enriched in NFE (2.83×10^{-4}) and essentially absent in most non-European populations. The truncating c.198_199del p.(Ala68*) has exome AF 1.34×10^{-4}.

Finding 4 — KO models recapitulate the biochemical defect; NIT1 is conserved bacteria→human

"Nit1-KO mutants of both human and yeast cells accumulate dGSH and the same compound is excreted in large amounts in the urine of Nit1-KO mice" (PMID: 28373563). Homologs function as dGSH amidases in *E. coli*, other GSH-producing bacteria, and plants; *"Ablating the Arabidopsis Nit1 gene causes a massive accumulation of dGSH"* (PMID: 30692244). Independent Nit1−/− mouse lines show a cancer-prone phenotype additive with Fhit loss (PMID: 19479888) and T-cell negative regulation (PMID: 19395373) — functionally distinct contexts from the cerebrovascular disease.

Finding 5 — ClinVar expands the allele spectrum; p.Arg243Trp has conflicting classifications and most variants are VUS

ClinVar lists additional Pathogenic BSVD4 alleles (c.198_199del p.(Ala68*); OMIM allelic variants "NIT1, 457G-A" and c.670dup). The recurrent p.Arg243Trp is "Conflicting classifications of pathogenicity," consistent with its relatively high NFE frequency. Most other NIT1 coding variants are VUS. Curation caution: NIT1 c.591+2T>G is Likely pathogenic only for severe-COVID-19 susceptibility, not BSVD4; adjacent DEDD-gene variants can appear under NIT1 queries.

Mechanistic Model / Interpretation

The disease is best understood as a **glutathione metabolite-repair deficiency manifesting as cerebral microangiopathy**. The upstream lesion (NIT1 LoF → dGSH accumulation) is firmly demonstrated at protein, enzyme, and model-organism levels. The downstream vascular pathology (electron-dense arterial-wall deposits → dilated perivascular spaces, hemorrhage, movement disorders) is well documented in patients, but the **causal link between dGSH accumulation and vascular deposition remains inferred** — the chemical nature of the deposits and the pathway by which a soluble metabolite damage product produces mural deposits has not been mechanistically established. This is the central open question.

NIT1-SVD takes its place alongside other monogenic cSVDs (CADASIL/NOTCH3, CARASIL/HTRA1, COL4A1/2, RVCL-S/TREX1, Fabry/GLA) but is mechanistically distinct: it is the first cSVD attributed to a **glutathione-pathway metabolite-repair enzyme**, and uniquely recessive

among the classic monogenic cSVDs. The dramatically dilated basal ganglia perivascular spaces — reflecting impaired perivascular/glymphatic clearance — are an unusually prominent imaging signature.

Evidence Base

PMID	Role	How it supports/challenges findings
38430071	**Defining clinical paper** (human cohort)	Establishes the disease, triad, inheritance, variants, LoF mechanism (absent protein, lost enzyme activity), autopsy deposits. Primary source for Findings 1, 3, 5.
28373563	Enzymology / model organisms	Identifies NIT1 as dGSH-hydrolyzing metabolite-repair enzyme; KO models accumulate/excrete dGSH; conservation to bacteria. Findings 2, 4.
30692244	Biochemistry / plant model	dGSH is an aminotransferase-generated damage product; Nit1 is dual-targeted amidase; Arabidopsis KO accumulates dGSH. Findings 2, 4.
19479888	Mouse model (tumor context)	Nit1−/− mice; cytoplasm+mitochondria localization; distinct Fhit-related function. Finding 4 / subcellular localization.
19395373	Mouse model (immune context)	Nit1 as T-cell negative regulator — a separate functional context. Finding 4.
42771568	Review — hereditary cSVD	Context for differential diagnosis and supportive management principles.
41456992	Population imaging study	Vascular risk factors associate with basal-ganglia perivascular spaces (modifier plausibility, not NIT1-specific).

Limitations and Knowledge Gaps

1. **Single small cohort (n=7).** All disease-specific clinical knowledge derives from one 2024 report. Prevalence, penetrance, expressivity, sex ratio, and natural-history timelines cannot be reliably quantified.

2. **Causal gap (dGSH → vascular deposits).** The step linking metabolite accumulation to arterial-wall electron-dense deposits is inferred; the composition of the deposits is unresolved.
3. **No cerebrovascular animal model.** Existing Nit1-KO models show the biochemical defect but not the brain microangiopathy.
4. **Variant interpretation.** p.Arg243Trp has conflicting ClinVar classifications; most other NIT1 variants are VUS — functional assays are needed.
5. **No treatment evidence.** No therapy has been tested; management guidance is extrapolated from other cSVDs.
6. **Curation hazards.** NIT1 c.591+2T>G (COVID-19 susceptibility) and neighboring DEDD-gene variants can be mis-attributed to BSVD4.

Proposed Follow-up Experiments / Actions

1. **Establish a dGSH biomarker assay** (plasma/urine/CSF LC-MS) as a diagnostic and potential therapy-monitoring endpoint; validate against genotype.
2. **Generate a brain-vascular Nit1 model** (conditional endothelial or smooth-muscle Nit1 knockout; patient-iPSC-derived cerebral vascular organoids) to test whether dGSH accumulation reproduces mural deposits and perivascular-space dilatation.
3. **Characterize the vascular deposits** biochemically (proteomics/metabolomics of autopsy vessel walls) to define the dGSH→deposit mechanism.
4. **Functional reclassification of NIT1 variants** via enzyme-activity and protein-stability assays to resolve p.Arg243Trp and reclassify VUS (ClinGen-style).
5. **Build an international registry / matchmaking effort** (GeneMatcher) to expand the cohort, refine penetrance/natural history, and assess vascular-risk-factor modifiers.
6. **Explore substrate-reduction / enzyme-replacement concepts** preclinically, given the loss-of-function enzyme-deficiency etiology.
7. **Issue curation guidance** flagging the COVID-19 variant and DEDD-locus pitfalls for knowledge-base integrity.

Report compiled from a 5-iteration autonomous investigation. Primary disease-specific evidence: Rutten et al., 2024 (PMID: 38430071). Evidence-source types are indicated throughout (human clinical, model organism, in vitro, computational/population-genetics).
